G H Raisoni College of Engineering and Management, Jalgaon

Department of CSE (Artificial Intelligence and Machine Learning)

PROJECT WEEKLY REPORT

Project Name: ContextBasedRAG: A Context-Aware Retrieval-Augmented Generation Assistant for Factory Maintenance

Work done during week 1: From Date: 31/08/2026 **To Date:** 06/09/2026

1. Studied the problem: maintenance knowledge in small factories is scattered across equipment manuals, SOPs and breakdown registers; defined the problem statement and objectives.
2. Literature survey of Retrieval-Augmented Generation (Lewis et al., REALM, Fusion-in-Decoder) and the RAG survey by Gao et al.
3. Studied dense vector search vs BM25 keyword ranking, Reciprocal Rank Fusion, hallucination and citation studies, and prompt-injection risks.
4. Wrote the functional and non-functional requirements of the system.
5. Carried out the feasibility study (technical, operational, economic, legal) and risk analysis.

Group Members :

1. Hitesh Hemkant Ingale (24213005)
2. Harshal Sanjay Pawar (24213003)
3. Harshal Mukesh Joshi (23113010)
4. Kunal Ravindra Dhokhane (23113005)

Guide Name & Sign **HOD**

Dr. Swati Patil

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Work done during week 2: From Date: 07/09/2026 **To Date:** 13/09/2026

1. Designed the system architecture: web frontend, FastAPI backend, MongoDB, ChromaDB vector store and Google Gemini.
2. Drew the Level 0 and Level 1 data flow diagrams.
3. Drew the UML diagrams: use case, sequence, activity and class diagrams.
4. Designed the work-order state diagram (draft, pending approval, approved, in progress, completed).
5. Finalised hardware and software requirements and the technology stack (Python, FastAPI, MongoDB, ChromaDB, Gemini).

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Work done during week 3: From Date: 14/09/2026 **To Date:** 20/09/2026

1. Built document upload with file validation for PDF, DOCX and PPTX files.
2. Implemented page-aware chunking so every passage keeps its page number.
3. Generated Gemini embeddings and stored them in the ChromaDB vector store.
4. Implemented BM25 keyword ranking and merged it with vector search using Reciprocal Rank Fusion (top 6 passages).
5. Started the single-page web interface in HTML, CSS and JavaScript.

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Work done during week 4: From Date: 21/09/2026 **To Date:** 27/09/2026

1. Gemini writes answers only from the numbered evidence passages.
2. Citations are validated by backend code; source names and pages are filled in from stored data, not from the model.
3. Replies "not found" when no evidence supports an answer; Hindi and Hinglish questions supported.
4. Login and four roles: viewer, technician, maintenance manager, administrator.
5. Isolation between factories: every record and chunk carries its factory id and every query filters on it.

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Work done during week 5: From Date: 28/09/2026 **To Date:** 04/10/2026

1. Asset register: create, search, edit and archive machines.
2. Breakdown records with CSV import and a per-row error report.
3. Repeat-issue summary computed from the breakdown records.
4. Work orders with manager approval (draft, approved, in progress, completed).
5. Audit trail recording every action with who, when and what.

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Work done during week 6: From Date: 05/10/2026 **To Date:** 11/10/2026

1. Black-box test cases TC1 to TC12: invalid file, cited answer, conflicting values, refusal, cross-factory question, prompt injection, approvals, CSV import, password limit, Hinglish question.
2. White-box and unit tests (tokenisation, BM25 and RRF order, work-order transitions, unit conversion); automated suite on in-memory MongoDB with a fake model.
3. Prepared 26 labelled evaluation questions; manual testing on desktop and mobile browsers.
4. Put the code on GitHub and added the Render deployment blueprint (6 Oct).
5. Fixed bugs found in testing: upload crash, mobile layout, vector-store 500 errors; switched embeddings to the Gemini API.

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Work done during week 7: From Date: 12/10/2026 **To Date:** 18/10/2026

1. Rebuilt as a factory maintenance assistant; conflict detector evaluated on 22 labelled cases and fixed (11 of 11 conflicts found, 0 false alarms).
2. Evaluated 26 questions: all labelled sources retrieved in the top 3 passages and every unanswerable question refused; demo deployed live on Render.
3. Added per-user access control, Excel/CSV documents and Excel export of registers.
4. Added Razorpay registration fee (test mode) with emails, reliability dashboard (MTTR, MTBF, downtime), forgot password and PDF bills.
5. All 137 automated tests passed (15 Oct); project report drafted.

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Work done during week 8: From Date: 19/10/2026 **To Date:** 25/10/2026

1. Final review of the project report with the guide and corrections.
2. Preparation of the presentation slides and demonstration script.
3. Final check of the live deployment with the demo factory data.
4. Submission of the project report.
5. Project presentation and live demonstration.

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Dr. Swati Patil